Guilherme Ribeiro Figueira

C++/C# Game Programmer (Junior)

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Experience

Unity XR Developer - Curricular Internship 03/2023 - 08/2023
Glartek Leiria, PT (Hybrid)

- Implemented an AR 3D model manipulation feature for **Microsoft Hololens 2** using Unity and MRTK while collaborating with the web development and design teams, adding to the project's feature suit.
- Utilized **3D vector math**, **memoization techniques**, and **search algorithms** to optimize performance and ensure reliable processing of imported 3D model for the supported formats.
- Created **unit and integration tests** for scripts using **Zenject dependency injection** framework, improving test coverage and code reliability.
- Collaborated using an **AGILE** methodology with **GitLab** for version control.

Education

Nova School Of Science and Technology Lisbon, PT
B.Sc. + M.Sc. in Computer Science and Engineering 2020 - 2025

- **Master Thesis**: *Dynamic Substitutional Reality In Adaptive Virtual Experiences*. Researched and developed a Unity-based framework that enables the abstract design of SR experiences, allowing them to be dynamically adapted to arbitrary physical spaces.
- Co-authored an additional paper P1.
- Part of my research and implementation is being ported into a **real-world museum application** at the *Portuguese National Tile Museum* in Lisbon, Portugal.
- **Relevant courses**: Games and Simulations, Computer Graphics, Object Oriented Programming, AI, Concurrent Programming, Computer Architecture, Operating Systems.

Skills

Programming Languages:	C++, C#, Java, Python, OCaml, HLSL
Engines & Frameworks:	Unreal Engine 5, Unity, FMOD, ImGui, SDL2
Graphics & Tools:	OpenGL, WebGL, Git, Shell, VIM, Docker
Creative Software:	Blender (Basics), Photoshop, DaVinci Resolve
Languages:	Portuguese (native), English (proficient)

Publications

P1 **Towards Augmented XR Movement and Interaction Confidence across Substitutional Reality**
Guilherme R. Figueira, Rui Nóbrega
International Conference on Graphics and Interaction (ICGI), 2025

Projects

The Shape of Silence - Unreal Engine 5 C++ Game 01/2024 - Present
Personal Project

- A narrative first person puzzle I started developing to sharpen my Unreal Engine C++ skills. I am working on this project solo which means doing all the game design, writing, sound design, and programming.
- So far I have developed an **extandable grid-based puzzle mechanic**, an **inventory subsystem** with support for item inspections, and a **dialogue system** using my own **EasyDialogueEditor** tool to write the dialogue.

OpenGL 3D renderer 07/2025 - Present
Personal Project

- Used **C++** and **OpenGL** to implement a decoupled renderer, seperating the rendering, application, shaders and settings.
- Finished features include: **3D fly-style camera**, **textured 3D objects**, a **fog setting**, and a **settings panel** added with **ImGui** to control camera, window and fog settings.
- On-going project, with the goal of eventually growing into my own small 3D game engine

Easy Dialogue Editor - C++ Desktop Application 03/2025 (3 weeks)
Personal Project

- A lightweight and platform agnostic **desktop program for editing game dialogue trees**. Created to solve a problem in a personal UE5 project.
- Written in **C++** with **ImGui**. It compiles to a **single small and portable executable** (~2.7 Mb) with no external dependancies, everything is statically linked.

The Natural's Descent - Unity Game 04/2024 - 06/2024
University Games and Simulations Course

- Action-adventure puzzle game developed in Unity as part of a **2-person team**.
- Designed and implemented a **procedurally generated cavern** using the **Wave Function Collapse** algorithm adapted to 3D, ensuring natural looking layouts with procedurally spawned enemies.
- Programmed multiple core gameplay mechanics such as: an **extandable sigil puzzle** component, a **ridable vehicle** and **FPS combat with AI animated humanoid enemies**.